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Thermodynamic evaluation of the Co–Sc and Fe–Sc systems

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Abstract

The thermodynamic assessments of the Co–Sc and Fe–Sc binary systems were carried out by using the CALPHAD (Calculation of Phase Diagrams) method on the basis of the experimental data including the thermodynamic properties and phase equilibria. The Gibbs free energies of the liquid, bcc, fcc, and hcp phases were described by the subregular solution model with the Redlich–Kister equation, and those of the intermetallic compounds (Co_2Sc, CoSc, $CoSc_2$, $CoSc_3$, αFe_2Sc, βFe_2Sc, λFe_2Sc and Fe_6Sc_{29}) in these two binary systems were described by the sublattice model. The thermodynamic parameters of the Co–Sc and Fe–Sc binary systems were optimized for reproducing the experimental data on phase diagrams and thermodynamic properties, and an agreement between the calculated results and experimental data was obtained in each binary system.

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Keywords: Phase diagrams; Co-based alloys and compounds; Thermodynamic modeling

1. Introduction

Co-based alloys have important advantage applications in many fields, such as magnetic materials and potential high temperature materials [1,2]. Especially, the class of Co-based compounds with rare earth elements is of particular interest due to their outstanding magnetic properties [3]. The phase equilibria in Co-based alloy systems are of importance for design of new Co-based alloys. The CALPHAD method, which is a powerful tool to cut down on cost and time during development of materials [4], effectively provides a clear guideline for materials design. As a result, it is of great essence to establish the thermodynamic database for the Co-based alloy systems. In the present paper, as a part of the thermodynamic database of the Co-based alloys, the thermodynamic descriptions for the phase diagrams in the Co–Sc and Fe–Sc systems were carried out by combining with the experimental data in the frame work of the CALPHAD method.

2. Experimental information

2.1. The Co–Sc system

The Co–Sc system is consisted of four solution phases (αCo, εCo, αSc and βSc phases), and four intermetallic compounds (Co_2Sc, CoSc, $CoSc_2$, and $CoSc_3$ phases). The phase diagram in the Co–Sc system was originally proposed by Markiv et al. [5], who investigated the phase equilibria in compositional range from 0 to 80 at.% Sc, then was reviewed by Moffatt [6]. The phase diagram of the Co–Sc system reviewed by Okamoto [7] is shown in Fig. 1.

In addition, Selhaoui and Kleppa [8] determined the standard enthalpies of formation of the Co_2Sc, $CoSc_2$ phases at 1473 K with the reference states of pure αCo and pure αSc phases by direct synthesis calorimetry. And Esin et al. [9] investigated the enthalpy of mixing of liquid phase at 15 at.% Sc at 1873 K in the Co–Sc system.

2.2. The Fe–Sc system

The Fe–Sc system is consisted of five solution phases (αFe, γFe, δFe, αSc and βSc phases), and four intermetallic compounds (αFe_2Sc, βFe_2Sc, λFe_2Sc and Fe_6Sc_{29} phases). Many studies have focused on the phase equilibria in the Fe–Sc binary system. The investigation of the phase diagram in the Fe–Sc system was originally carried out by Naumkin et al. [10], where the solubility of Sc in the αFe phase is less than 0.5 at.% [11]. Then on the basis of the investigation of the phase diagram in compositional range from 0 to 90 at.% Sc using X-ray diffraction and different thermal analyses, Bodak et al. [12] found polymor-

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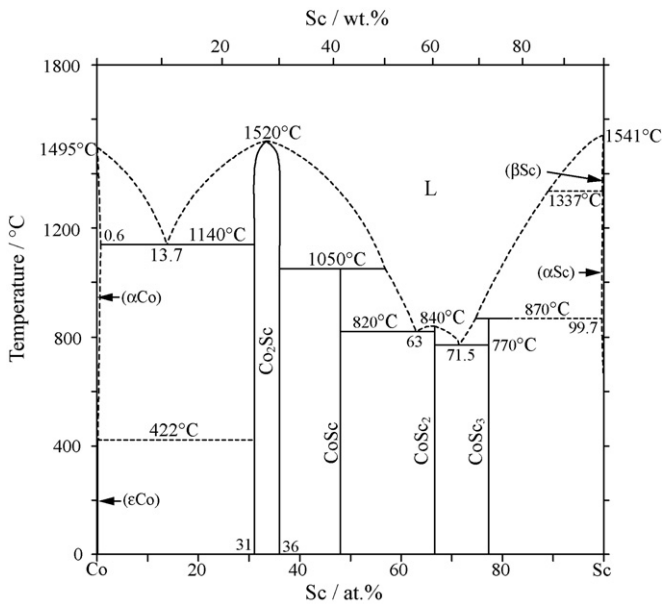

Fig. 1. The phase diagram of the Co–Sc system reviewed by Okamoto [7].

phic transformation of the Fe_2Sc phase, and determined the eutectic reaction (L ↔ $\alpha Fe_2Sc + \alpha Sc$, at 975 °C), and the formula of the $FeSc_7$ phases instead of the $FeSc_3$ phase. However, in subsequent studies, the formation of the compound of the approximate composition $FeSc_7$ was defined as the composition Fe_6Sc_{29} [13]. The phase diagram in the Fe–Sc system reviewed by Okamoto [7] is shown in Fig. 2.

Furthermore, Selhaoui and Kleppa [8] determined the standard enthalpy of formation of the Fe_2Sc phase at 1473 K with the reference states of pure αFe and pure αSc phases by direct synthesis calorimetry, and Goncharuk and Sidorko [14] measured those of the Fe_2Sc and Fe_6Sc_{29} phases at 950 K by using EMF later. However, there is the divergence between the results determined by Selhaoui and Kleppa [8] and Goncharuk and Sidorko [14], which need to be further studied. The enthalpies of mixing of liquid phase in compositional range from 0 to 18 at.% Sc at 1873 K in the Fe–Sc system were measured by Esin et al. [15]. Moreover, the Curie temperature of the αFe_2Sc phase was reported to be 269 °C [16].

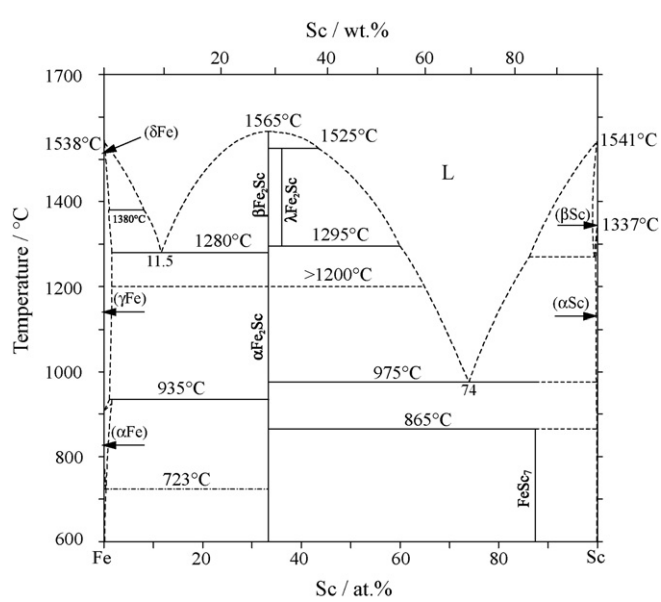

Fig. 2. The phase diagram of the Fe–Sc system reviewed by Okamoto [7].

Table 1
Stable solid phases and the used models in the Co–Sc [16] and Fe–Sc [7] systems

System	Phase	Strukturbericht designation	Prototype	Modeling phase	Used models
Co–Sc	(εCo)	A3	Mg	HCP	SSM
	(αCo)	A1	Cu	FCC	SSM
	Co_2Sc	C15	Cu_2Mg	$(Co,Sc)_2(Co,Sc)$	SM
	CoSc	B2	CsCl	(Co)(Sc)	SM
	$CoSc_2$	C16	Al_2Cu	$(Co)_1(Sc)_2$	SM
	$CoSc_3$			$(Co)(Sc)_3$	SM
	(βSc)	A2	W	BCC	SSM
	(αSc)	A3	Mg	HCP	SSM
Fe–Sc	(δFe)	A2	W	BCC	SSM
	(γFe)	A1	Cu	FCC	SSM
	(αFe)	A2	W	BCC	SSM
	βFe_2Sc	C36	$MgNi_2$	$(Fe)_2(Sc)_1$	SM
	αFe_2Sc	C14	$MgZn_2$	$(Fe)_2(Sc)_1$	SM
	λFe_2Sc	C15	Cu_2Mg	$(Fe)_2(Sc)_1$	SM
	Fe_6Sc_{29}			$(Fe)_{6(}Sc)_{29}$	SM
	(βSc)	A2	W	BCC	SSM
	(αSc)	A3	Mg	FCC	SSM

SSM, subregular solution model; and SM, sublattice model.

3. Thermodynamic models

The information of stable solid phases and the used models in the Co–Sc [17] and Fe–Sc [8] systems is listed in Table 1.

3.1. Solution phases

The Gibbs free energies of the liquid, bcc, fcc, and hcp phases in the Me–Sc (Me: Co, Fe) binary system are described by the subregular solution model, as follows:

$$G_m^\phi = \sum_{i=\mathrm{Me,Sc}} {}^0G_i^\phi x_i + \mathrm{RT} \sum_{i=\mathrm{Me,Sc}} x_i \ln x_i + \Delta^E G_m^\phi + \Delta^{\mathrm{mag}} G_m^\phi \quad (1)$$

where ${}^0G_i^\phi$ is the Gibbs free energy of pure component i in the respective reference state with the ϕ phase, which is taken from the SGTE database [18], R the gas constant, T the absolute temperature, x_i the molar fraction of the component i. The excess Gibbs free energy, $\Delta^E G_m^\phi$ is expressed in the Redlich–Kister polynomial [19] by

$$\Delta^E G_m^\phi = x_{\mathrm{Me}} x_{\mathrm{Sc}} \sum_{m=0}^{n} {}^m L_{\mathrm{Me,Sc}}^\phi (x_{\mathrm{Me}} - x_{\mathrm{Sc}})^m \quad (2)$$

where ${}^m L_{\mathrm{Me,Sc}}^\phi$ is the binary interaction parameter, which is expressed as follows:

$${}^m L_{\mathrm{Me,Sc}}^\phi = a + bT \quad (3)$$

where the parameters of a and b are evaluated on the basis of the experimental data in this work. $\Delta^{\mathrm{mag}} G_m^\phi$ is the magnetic contribution to the Gibbs free energy, which is described by the following equation [20]:

$$\Delta^{\mathrm{mag}} G_m^\phi = \mathrm{RT} \ln(\beta + 1) f(\tau) \quad (4)$$

where $\tau = T/T_C^{\phi}$, T_C^{ϕ} is the Curie temperature of solution for ferromagnetic ordering and β is the Bothr magneton number, the function $f(\tau)$ is formulated by the polynomial of the normalized temperature, as follows:

$$f(\tau) = 1 - \frac{[79\tau^{-1}/140p + 474/497(1/p - 1)(\tau^3/6 + \tau^9/135 + \tau^{15}/600)]}{D}, \quad \text{for } \tau \le 1 \tag{5}$$

$$f(\tau) = -\frac{(\tau^{-5}/10 + \tau^{-15}/315 + \tau^{-25}/1500)}{D}, \quad \text{for } \tau > 1 \tag{6}$$

$$D = \frac{518}{1125} + \frac{11,692}{15,975}\left(\frac{1}{p} - 1\right) \tag{7}$$

where p depends on the structure, 0.40 for bcc structure and 0.28 for others.

3.2. Stoichiometric intermetallic compounds

The intermetallic compounds of the CoSc, $CoSc_2$, $CoSc_3$, αFe_2Sc, βFe_2Sc, λFe_2Sc and Fe_6Sc_{29} phases in the Me–Sc (Me: Co, Fe) binary system are stoichiometric phases. The Gibbs free energy per molar of formula unit Me_pSc_q can be expressed, as follows:

$$G_{Me:Sc}^{Me_pSc_q} = p^0G_{Me}^{SER} + q^0G_{Sc}^{SER} + \Delta^0G_f^{Me_pSc_q} + \Delta^{mg}G_{Me:Sc}^{Me_pSc_q} \tag{8}$$

where the $\Delta^0G_f^{Me_pSc_q}$ indicates the standard Gibbs free energy of formation of the stoichiometric compound from pure elements at temperature T, which is described as

$$\Delta^0G_f^{Me_pSc_q} = a' + b'T + c'T\ln(T) \tag{9}$$

where the parameters of a', b' and c' are evaluated in the present work.

$\Delta^{mg}G_{Me:Sc}^{Me_pSc_q}$ is the magnetic contribution to $G_{Me:Sc}^{Me_pSc_q}$, which is expressed by Eqs. (4)–(7), where β and T_C^{ϕ} of each intermetallic compound are taken from the literature [16].

3.3. Intermetallic compound with solubility

Since the C15 Lavers phase (Co_2Sc) has solubility form 31 to 36 at.% Sc in the Co–Sc binary system, the appropriate sublattice model for this phase is adapted as $(Co, Sc)_{0.67}$:$(Co, Sc)_{0.33}$. The Gibbs free energy for one mole of formula unit of the Co_2Sc phase is expressed, as follows:

$$\begin{aligned} G^{Co_2Sc} = {} & y_{Co}^1y_{Sc}^2\,{}^0G_{Co:Sc}^{Co_2Sc} + y_{Co}^1y_{Co}^2\,{}^0G_{Co:Co}^{Co_2Sc} + y_{Sc}^1y_{Sc}^2\,{}^0G_{Sc:Sc}^{Co_2Sc} \\ & + y_{Sc}^1y_{Co}^2\,{}^0G_{Sc:Co}^{Co_2Sc} + 0.67\,RT(y_{Co}^1\ln y_{Co}^1 + y_{Sc}^1\ln y_{Sc}^1) \\ & + 0.33\,RT(y_{Co}^2\ln y_{Co}^2 + y_{Sc}^2\ln y_{Sc}^2) \\ & + y_{Co}^1y_{Sc}^1y_{Co}^2L_{Co,Sc:Co}^{Co_2Sc} + y_{Co}^1y_{Sc}^1y_{Sc}^2L_{Co,Sc:Sc}^{Co_2Sc} \\ & + y_{Co}^1y_{Co}^2y_{Sc}^2L_{Co:Co,Sc}^{Co_2Sc} + y_{Sc}^1y_{Co}^2y_{Sc}^2L_{Sc:Co,Sc}^{Co_2Sc} \end{aligned} \tag{10}$$

where y_i^1 and y_i^2 are the site fractions of component i in the first and second sublattices, respectively. The ${}^0G_{i:j}^{Co_2Sc}$ is expressed by

$${}^0G_{i:j}^{Co_2Sc} = 0.67\,{}^0G_i^{Co_2Sc} + 0.33\,{}^0G_j^{Co_2Sc} + a'' + b''T \tag{11}$$

$L_{i,j:k}^{Co_2Sc}$ is the binary interaction parameter, which is expressed as follows:

$$L_{i,j:k}^{Co_2Sc} = a''' + b'''T + c'''T\ln T \tag{12}$$

The parameters of a'', b'', a''', b''', and c''' are evaluated on the basis of the experimental data in this work.

The Co_2Sc phase may be considered as a Wanger–Schottky phase [21]. In such a case the following constraint among the ${}^0G_{i:j}^{Co_2Sc}$ parameters of Eq. (10) is used [22]:

$${}^0G_{Sc:Co}^{Co_2Sc} = {}^0G_{Co:Co}^{Co_2Sc} + {}^0G_{Sc:Sc}^{Co_2Sc} - {}^0G_{Co:Sc}^{Co_2Sc} \tag{13}$$

4. Optimized results and discussion

The optimization of the thermodynamic parameters was carried out by using the PARROT program in the Thermo-Calc software [23], which can handle various kinds of experimental data. The experimental data of the phase diagram and thermodynamic properties were used as input to the program. Each piece of selected information was given a certain weight based on the importance of data, and changed by trial and error during the assessment, until most of the selected experimental information was reproduced within the expected uncertainty limits.

4.1. The Co–Sc system

The calculated Co–Sc phase diagram with the experimental data is shown in Fig. 3, it is seen that the calculated phase diagram is in agreement with the experimental data. However, there exists a slight discrepancy in the solubility of the Co_2Sc phase

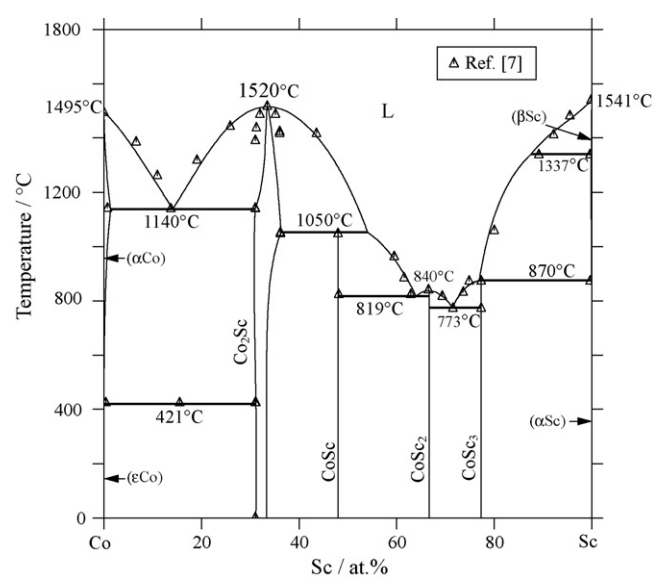

Fig. 3. The calculated phase diagram of the Co–Sc system with reaction temperatures compared with experimental data.

Table 2
Special points of the Co–Sc system

Invariant reaction	Reaction type	Composition Sc (at.%)			T (°C)	References
L ↔ βSc	Melting		100.0		1541	[7]
			100.0		1541	This work
βSc ↔ αSc	Polymorphic		100.0		1337	[7]
			100.0		1337	This work
L + αSc ↔ $CoSc_3$	Peritectic	–	99.7	77.3	870	[7]
		76.8	99.9	77.3	870	This work
L ↔ $CoSc_2$ + $CoSc_3$	Eutectic	71.5	66.7	77.3	770	[7]
		71.5	66.7	77.3	773	This work
L ↔ $CoSc_2$	Melting		66.7		840	[7]
			66.7		840	This work
L ↔ CoSc + $CoSc_2$	Eutectic	63.0	48.0	66.7	820	[7]
		63.9	48.0	66.7	819	This work
L + Co_2Sc ↔ CoSc	Peritectic	–	36.0	48.0	1050	[7]
		54.2	36.4	48.0	1050	This work
L ↔ Co_2Sc	Melting		31.0–36.0		1520	[7]
			31.1–33.3		1520	This work
L ↔ Co_2Sc + αCo	Eutectic	13.7	31.0	0.6	1140	[7]
		14.2	31.0	1.3	1140	This work
L ↔ αCo	Melting		0		1495	[7]
			0		1495	This work
αCo ↔ εCo	Polymorphic		0		422	[7]
			0		421	This work

between the calculated result and experimental data. The phase boundary of the Co_2Sc phase was determined by Markiv et al. [5] based on microstructural observation, but the experimental data are limited. The present calculated results indicate that the solubility of the Co_2Sc phase reduces with the fall of temperature, which accords with the thermodynamic rules. The special points in the Co–Sc system are listed in Table 2 compared with experimental data, where the different compositions of Sc correspond to those in the three phases in the order they appear in the invariant reactions. A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Co–Sc system are given in Table 3.

The calculated enthalpies of formation for the compounds compared with experimental data at 873 K with the reference states of pure αCo and pure αSc phases are presented in Fig. 4. And a comparison of enthalpies of mixing of liquid at 1873 K between calculated and experimental data is shown in Fig. 5. It is seen that a good agreement on the thermodynamic properties is obtained between the calculated results and experimental data.

4.2. The Fe–Sc system

In the Fe–Sc system, there are great differences between the experimental data reported by different researchers. In the present assessment, the latest experimental data [12,13] were selected for the optimization of thermodynamic parameters describing the eutectic reaction (L ↔ αFe_2Sc + αSc, at 975 °C) and the peritectoid reaction (αFe_2Sc + αSc ↔ Fe_6Sc_{29}, at 865 °C). The calculated Fe–Sc phase diagram with experimen-

Table 3
Thermodynamic parameters in the Co–Sc system assessed in the present work

Parameters in each phase (J/mol)
Liquid phase, format (Co, Sc)
${}^0L^{\mathrm{Liq}}_{\mathrm{Co,Sc}} = -155,000 + 30 \times T$
${}^1L^{\mathrm{Liq}}_{\mathrm{Co,Sc}} = 55,000 - 50 \times T$
${}^2L^{\mathrm{Liq}}_{\mathrm{Co,Sc}} = 50,000 - 20 \times T$
Fcc phase, format (Co, Sc)
${}^0L^{\mathrm{Fcc}}_{\mathrm{Co,Sc}} = -35,000$
${}^1L^{\mathrm{Fcc}}_{\mathrm{Co,Sc}} = -25,000$
Bcc phase, format (Co, Sc)
${}^0L^{\mathrm{Bcc}}_{\mathrm{Co,Sc}} = -9500$
Hcp phase, format (Co, Sc)
${}^0L^{\mathrm{Hcp}}_{\mathrm{Co,Sc}} = 20,000$
Co_2Sc phase, format $(Co, Sc)_{0.67}(Co, Sc)_{0.33}$
${}^0G^{\mathrm{Co_2Sc}}_{\mathrm{Co:Co}} = {}^0G^{\mathrm{hcp}}_{\mathrm{Co}} + 5000$
${}^0G^{\mathrm{Co_2Sc}}_{\mathrm{Sc:Sc}} = {}^0G^{\mathrm{hcp}}_{\mathrm{Sc}} + 5000$
${}^0G^{\mathrm{Co_2Sc}}_{\mathrm{Co:Sc}} = 0.67 \times {}^0G^{\mathrm{hcp}}_{\mathrm{Co}} + 0.33 \times {}^0G^{\mathrm{hcp}}_{\mathrm{Sc}} - 38566.67 + 2.00 \times T$
${}^0G^{\mathrm{Co_2Sc}}_{\mathrm{Sc:Co}} = {}^0G^{\mathrm{Co_2Sc}}_{\mathrm{Co:Co}} + {}^0G^{\mathrm{Co_2Sc}}_{\mathrm{Sc:Sc}} - {}^0G^{\mathrm{Co_2Sc}}_{\mathrm{Co:Sc}}$
${}^0L^{\mathrm{Co_2Sc}}_{\mathrm{Co,Sc:Sc}} = -5666.67$
${}^0L^{\mathrm{Co_2Sc}}_{\mathrm{Co:Co,Sc}} = -17678.67 + 53.17 \times T - 5.12 \times T \times \ln(T)$
CoSc phase, format $(Co)_{0.52}(Sc)_{0.48}$
${}^0G^{\mathrm{CoSc}}_{\mathrm{Co:Sc}} = 0.52 \times {}^0G^{\mathrm{hcp}}_{\mathrm{Co}} + 0.48 \times {}^0G^{\mathrm{hcp}}_{\mathrm{Sc}} - 43672.59 + 7.83 \times T$
$CoSc_2$ phase, format $(Co)_{0.33}(Sc)_{0.67}$
${}^0G^{\mathrm{CoSc_2}}_{\mathrm{Co:Sc}} = 0.33 \times {}^0G^{\mathrm{hcp}}_{\mathrm{Co}} + 0.67 \times {}^0G^{\mathrm{hcp}}_{\mathrm{Sc}} - 28611.67 + 2.00 \times T$
$CoSc_3$ phase, format $(Co)_{0.23}(Sc)_{0.77}$
${}^0G^{\mathrm{CoSc_3}}_{\mathrm{Co:Sc}} = 0.23 \times {}^0G^{\mathrm{hcp}}_{\mathrm{Co}} + 0.77 \times {}^0G^{\mathrm{hcp}}_{\mathrm{Sc}} - 20749.15 + 1.54 \times T$

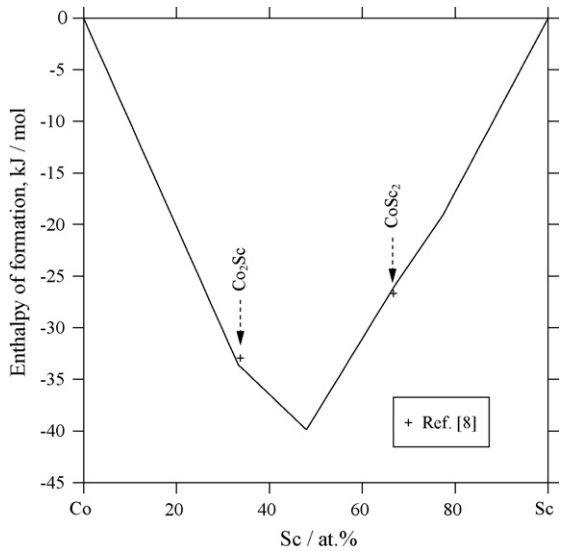

Fig. 4. Calculated enthalpy of formation in the Co–Sc system with experimental data at 873 K. Reference states: αCo (fcc) for Co and αSc (hcp) for Sc.

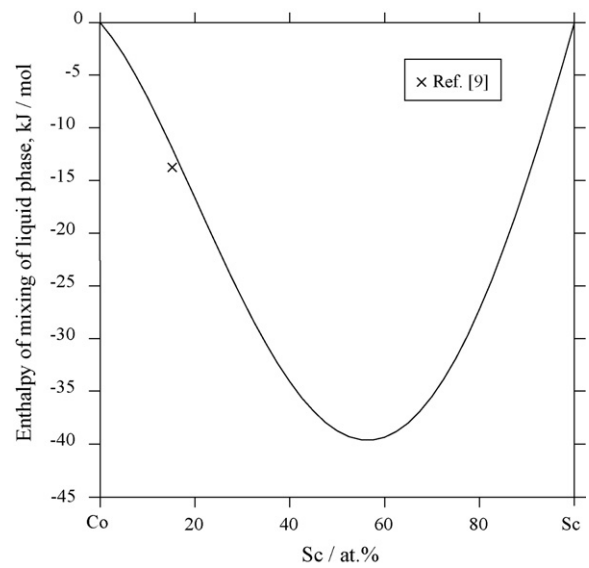

Fig. 5. Calculated enthalpies of mixing of liquid phase in the Co–Sc system with the experimental data at 1873 K. Reference states: liquid phases for Co and Sc.

Table 4
Special points of the Fe–Sc system

Invariant Reaction	Reaction type	Composition Sc / at.%			T (°C)	References
L ↔ δFe	Melting		100.0		1538	[7]
			100.0		1538	This work
δFe ↔ γFe	Allotropic		100.0		1412	[7]
			100.0		1412	This work
δFe ↔ γFe + L	Peritectic	–	–	–	1380	[7]
		1.2	1.3	6.5	1380	This work
γFe ↔ αFe	Allotropic		100.0		912	[7]
			100.0		912	This work
γFe + αFe_2Sc ↔ αFe	Peritectoid	–	33.3	–	935	[7]
		0.5	33.3	0.7	935	This work
L ↔ (γFe) + βFe_2Sc	Eutectic	11.5	–	33.3	1280	[7]
		10.2	3.0	33.3	1280	This work
L ↔ βFe_2Sc	Congruent		33.3		1565	[7]
			33.3		1565	This work
βFe_2Sc ↔ αFe_2Sc	Polymorphic		33.3		1200	[7]
			33.3		1200	This work
L + βFe_2Sc ↔ λFe_2Sc	Peritectic	–	33.3	35.7	1525	[7]
		44.2	33.3	35.7	1525	This work
L ↔ αFe_2Sc + αSc	Eutectic	74.0	33.3	–	975	[7]
		75.1	33.3	99.5	975	This work
αFe_2Sc + αSc ↔ Fe_6Sc_{29}	Peritectoid	33.3	–	87.5	865	[7]
		33.3	99.7	82.9	865	This work
L ↔ βSc	Melting		0		1541	[7]
			0		1541	This work
βSc ↔ αSc	Polymorphic		0		1337	[7]
			0		1337	This work
βSc ↔ L + αSc	Peritectic	–	–	–	–	[7]
		98.5	85.9	99.7	1271	This work

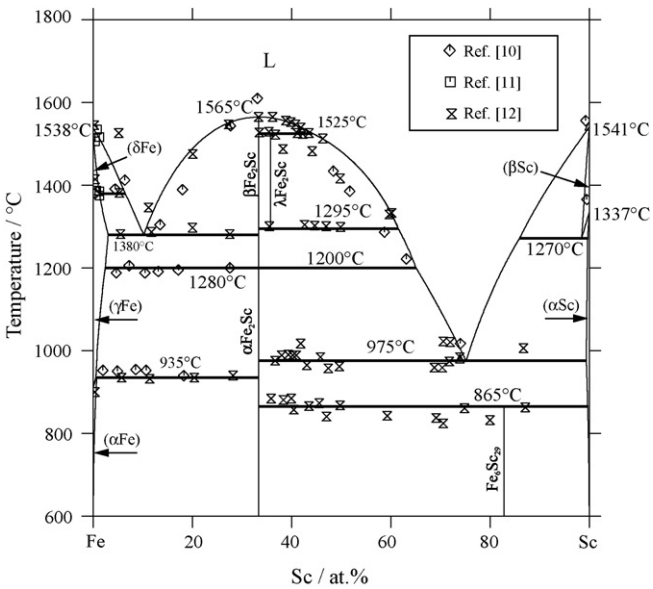

Fig. 6. The calculated phase diagram of the Fe–Sc system with reaction temperatures compared with experimental data.

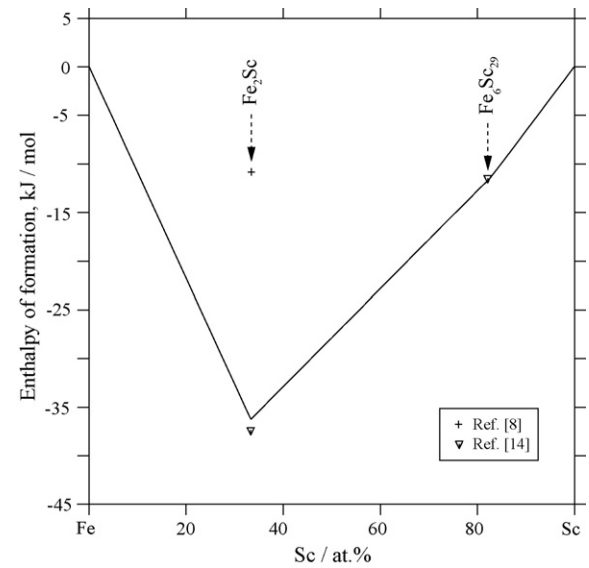

Fig. 7. Calculated enthalpy of formation in the Fe–Sc system with experimental data at 950 K. Reference states: αFe (bcc) for Fe and αSc (hcp) for Sc.

tal data is shown in Fig. 6. It is shown that the calculated results are in good agreement with experimental data. The calculated compositions and temperatures for the invariant reactions are listed in Table 4 with the selected experimental data [7]. A complete set of the thermodynamic parameters describing the Gibbs free energy of each phase in this system is given in Table 5.

A comparison of the enthalpies of formation for the compounds between calculated results and experimental data at 950 K with the reference states of pure γFe and pure αSc phases is shown in Fig. 7, it is seen that there is larger difference on the data of the Fe_2Sc phase reported by Selhaoui and Kleppa [8] and Goncharuk and Sidorko [14], and the calculated results are close to that reported by Goncharuk and Sidorko [14]. And the calculated enthalpies of mixing of liquid at 1873 K compared with experimental data are shown in Fig. 8. It is seen that an excellent agreement is obtained between the calculated results and experimental data.

Table 5
Thermodynamic parameters in the Fe–Sc system assessed in the present work

Parameters in each phase (J/mol)	References
Liquid phase, format (Fe, Sc)	
${}^{0}L^{\mathrm{Liq}}_{\mathrm{Fe,Sc}} = -55,000 - 5 \times T$	
${}^{1}L^{\mathrm{Liq}}_{\mathrm{Fe,Sc}} = 15,000 - 20 \times T$	
${}^{2}L^{\mathrm{Liq}}_{\mathrm{Fe,Sc}} = 15,000 - 20 \times T$	
Fcc phase, format (Fe, Sc)	
${}^{0}L^{\mathrm{Fcc}}_{\mathrm{Fe,Sc}} = -32571.5 - 9.775 \times T$	
Bcc phase, format (Fe, Sc)	
${}^{0}L^{\mathrm{Bcc}}_{\mathrm{Fe,Sc}} = -27,500$	
${}^{1}L^{\mathrm{Bcc}}_{\mathrm{Fe,Sc}} = -17,500$	
Hcp phase, format (Fe, Sc)	
${}^{0}L^{\mathrm{Hcp}}_{\mathrm{Fe,Sc}} = 5700$	
αFe_2Sc phase, format $(Fe)_{0.67}(Sc)_{0.33}$	
${}^{0}G^{\alpha Fe_2Sc}_{\mathrm{Fe:Sc}} = 0.67 \times {}^{0}G^{\mathrm{bcc}}_{\mathrm{Fe}} + 0.33 \times {}^{0}G^{\mathrm{hcp}}_{\mathrm{Sc}} - 40018.67 + 7.63 \times T$	
$T_C^{\alpha Fe_2Sc} = 542$	[16]
$\beta^{\alpha Fe_2Sc} = 1.37$	[16]
βFe_2Sc phase, format $(Fe)_{0.67}(Sc)_{0.33}$	
${}^{0}G^{\beta Fe_2Sc}_{\mathrm{Fe:Sc}} = 0.67 \times {}^{0}G^{\mathrm{bcc}}_{\mathrm{Fe}} + 0.33 \times {}^{0}G^{\mathrm{hcp}}_{\mathrm{Sc}} + 96,673 - 691.67 \times T + 83.14 \times T \times \ln(T)$	
λFe_2Sc phase, format $(Fe)_{0.64}(Sc)_{0.36}$	
${}^{0}G^{\lambda Fe_2Sc}_{\mathrm{Fe:Sc}} = 0.64 \times {}^{0}G^{\mathrm{bcc}}_{\mathrm{Fe}} + 0.36 \times {}^{0}G^{\mathrm{hcp}}_{\mathrm{Sc}} + 171262.86 - 1041.99 \times T + 124.34 \times T \times \ln(T)$	
Fe_6Sc_{29} phase, format $(Fe)_{0.17}(Sc)_{0.83}$	
${}^{0}G^{Fe_6Sc_{29}}_{\mathrm{Fe:Sc}} = 0.17 \times {}^{0}G^{\mathrm{bcc}}_{\mathrm{Fe}} + 0.83 \times {}^{0}G^{\mathrm{hcp}}_{\mathrm{Sc}} - 12285.71 + 3.69 \times T$	

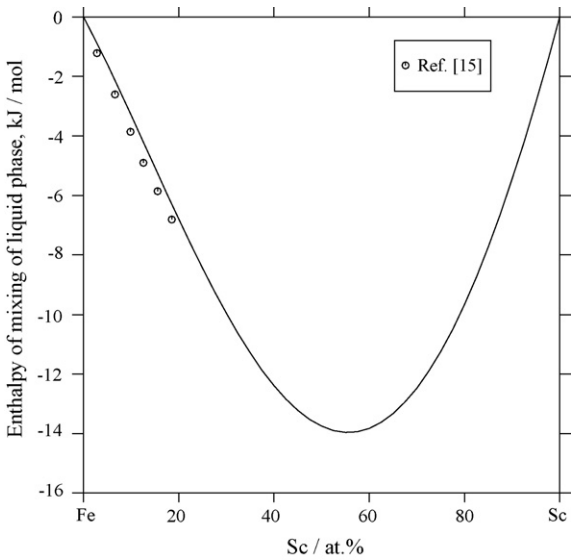

Fig. 8. Calculated enthalpies of mixing of liquid phase in the Fe–Sc system with experimental data at 1873 K. Reference states: liquid phases for Fe and Sc.

5. Conclusions

The Co–Sc and Fe–Sc binary systems were critically assessed by combining thermodynamic models with the experimental data including phase equilibria and thermodynamic data, and most of experimental information can be satisfactorily reproduced on the basis of the optimized thermodynamic parameters.

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